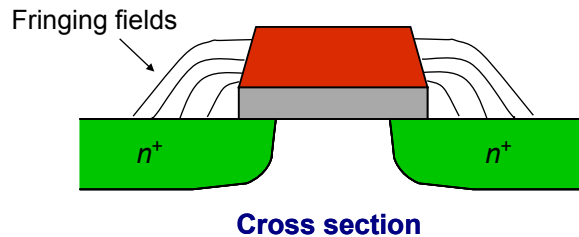


Gate Fringe Capacitance



- C_{OV} not just from metallurgic overlap – get fringing fields too
- Typical value: $\sim 0.2 \text{ fF} \cdot W (\text{in } \mu\text{m}) / \text{edge}$

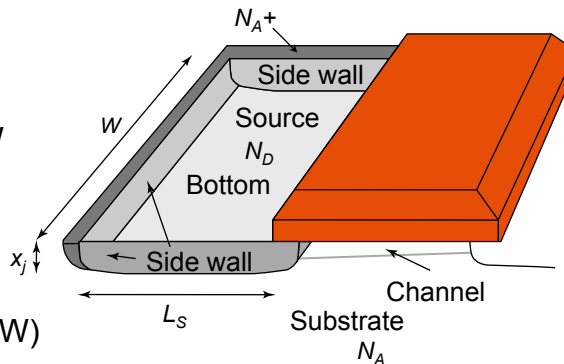
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Diffusion Capacitance

- Bottom
 - Area cap
 - $C_{\text{bottom}} = C_j \cdot L_S \cdot W$
- Sidewalls
 - Perimeter cap
 - $C_{\text{sw}} = C_{j\text{sw}} \cdot (2L_S + W)$
- GateEdge
 - $C_{\text{ge}} = C_{j\text{gate}} \cdot W$
 - Usually automatically included in the SPICE model



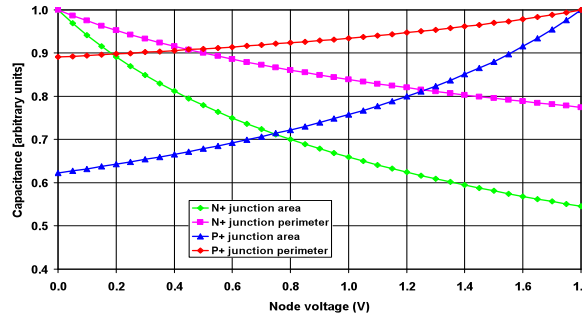
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Junction Capacitance (2)

- Junction caps are nonlinear
 - C_J is a function of junction bias



- SPICE model equations:

- Area $C_J = \text{area} \times C_{J0} / (1 + |V_{DB}|/\phi_B)^{m_j}$
- Perimeter $C_J = \text{perim} \times C_{JSW} / (1 + |V_{DB}|/\phi_B)^{m_{jsw}}$
- Gate edge $C_J = W \times C_{Jgate} / (1 + |V_{DB}|/\phi_B)^{m_{jswg}}$

- How do we deal with nonlinear capacitance?

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Linearizing the Junction Capacitance

Replace non-linear capacitance by
large-signal equivalent linear capacitance
 which displaces equal charge over voltage swing of interest

$$C_{eq} = \frac{\Delta Q_j}{\Delta V_D} = \frac{Q_j(V_{high}) - Q_j(V_{low})}{V_{high} - V_{low}} = K_{eq} C_{j0}$$

$$K_{eq} = \frac{-\phi_0^m}{(V_{high} - V_{low})(1 - m)} [(\phi_0 - V_{high})^{1-m} - (\phi_0 - V_{low})^{1-m}]$$

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Capacitance Model Summary

□ Gate-Channel Capacitance

- $C_{GC} \approx 0$ ($|V_{GS}| < |V_T|$)
- $C_{GC} = C_{ox} \cdot W \cdot L_{eff}$ (Linear)
 - 50% G to S, 50% G to D
- $C_{GC} = (2/3) \cdot C_{ox} \cdot W \cdot L_{eff}$ (Saturation)
 - 100% G to S

□ Gate Overlap Capacitance

- $C_{GSO} = C_{GDO} = C_O \cdot W$ (Always)

□ Junction/Diffusion Capacitance

- $C_{diff} = C_j \cdot L_S \cdot W + C_{jsw} \cdot (2L_S + W) + C_{jg}W$ (Always)

Capacitances in 0.25 μm CMOS Process

	C_{ox} (fF/ μm^2)	C_O (fF/ μm)	C_j (fF/ μm^2)	m_j	ϕ_b (V)	C_{jsw} (fF/ μm)	m_{jsw}	ϕ_{dsw} (V)
NMOS	6	0.31	2	0.5	0.9	0.28	0.44	0.9
PMOS	6	0.27	1.9	0.48	0.9	0.22	0.32	0.9

CMOS Inverter VTC

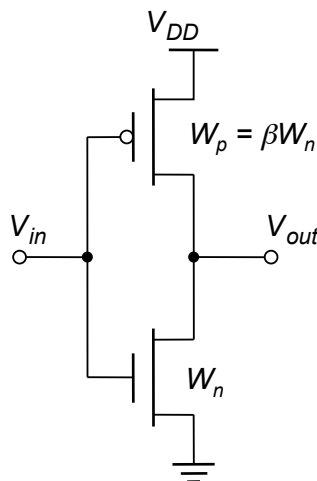


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The CMOS Inverter



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PMOS Load Lines

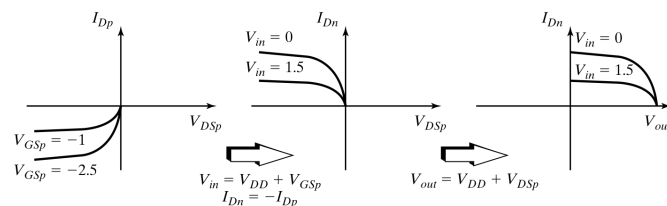
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PMOS Load Lines

- For DC VTC, $I_{Dn} = I_{Dp}$
 - Graphically, looking for intersections of NMOS and PMOS IV characteristics
- To put IV curves on the same plot, PMOS IV is “flipped” since $|V_{DSp}| = V_{DD} - V_{out}$
 - Also, $|V_{GS p}| = V_{dd} - V_{in}$

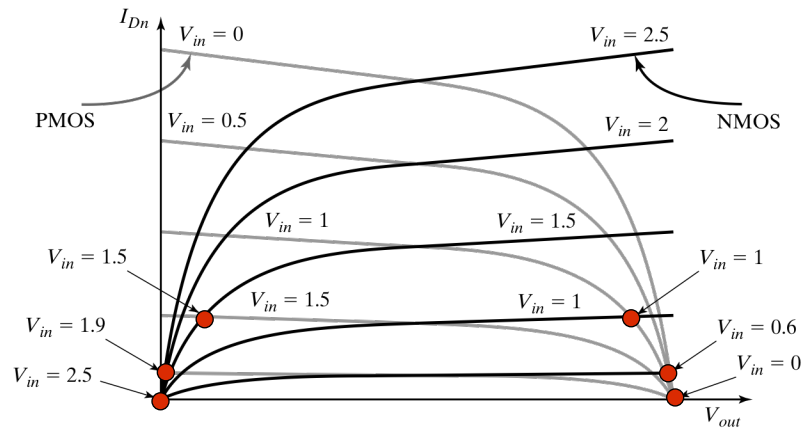


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CMOS Inverter Load Characteristics

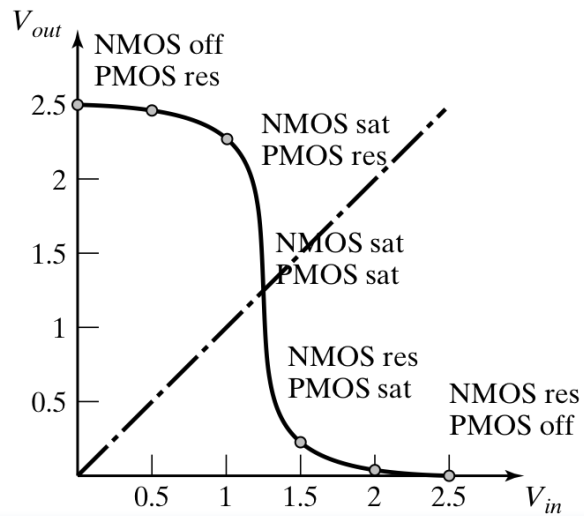


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CMOS Inverter VTC

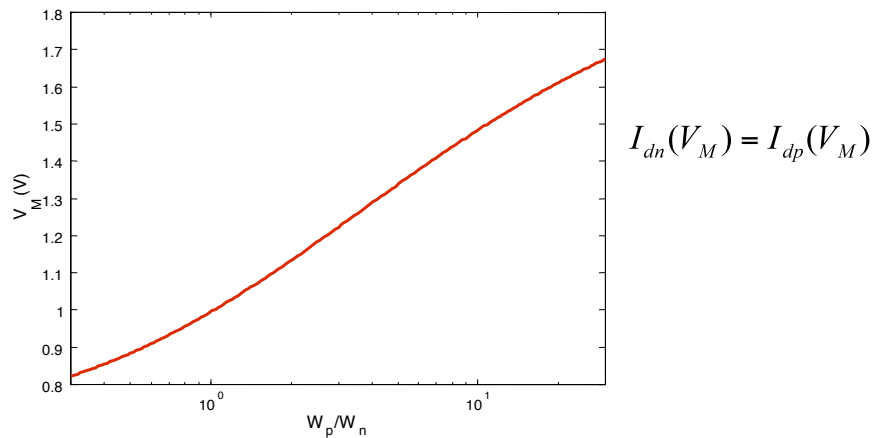


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Switching Threshold as a Function of Transistor Ratio

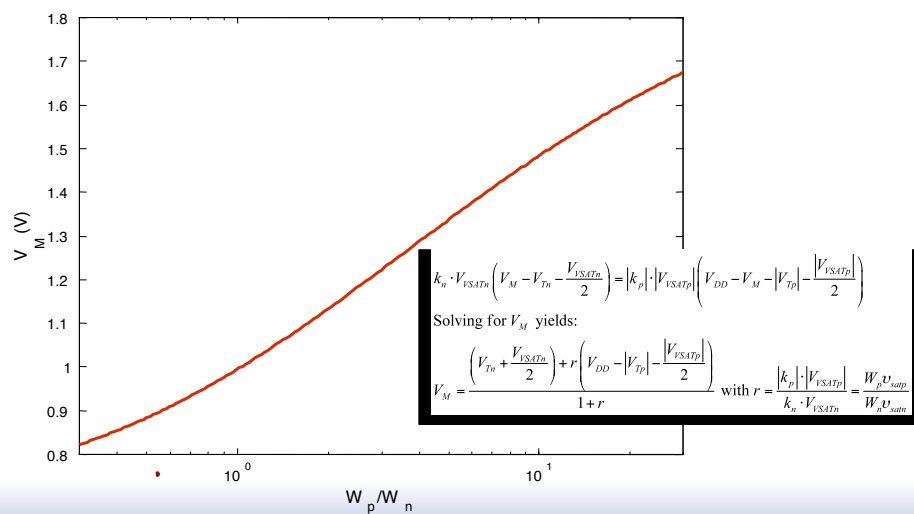


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Switching Threshold as a Function of Transistor Ratio



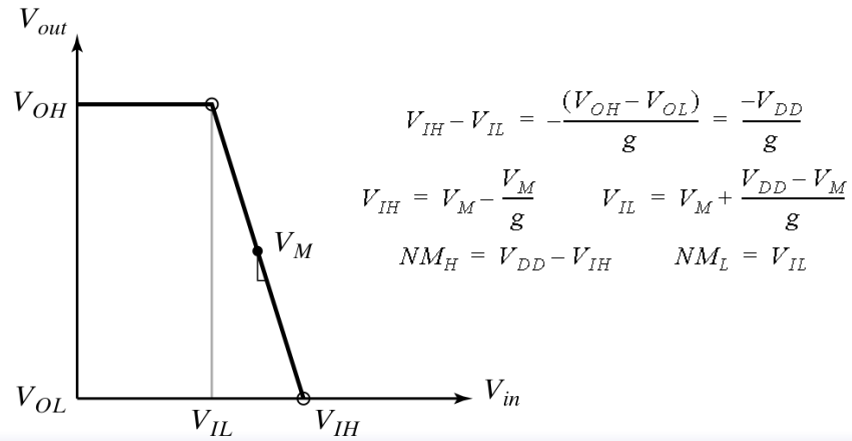
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Determining V_{IH} and V_{IL}

A simplified approach

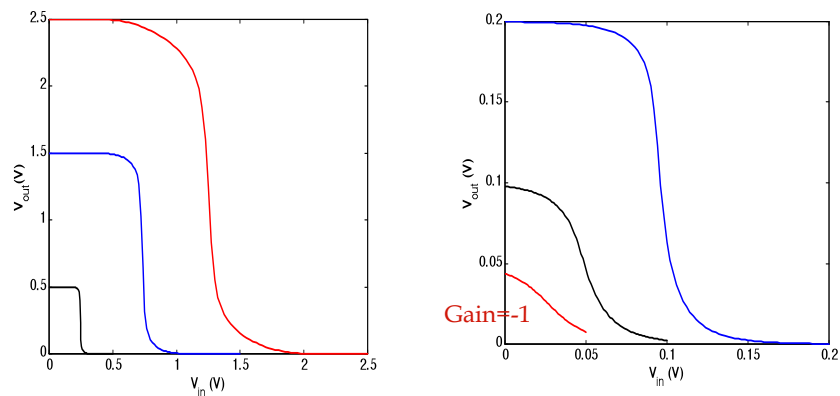


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Gain as a function of VDD

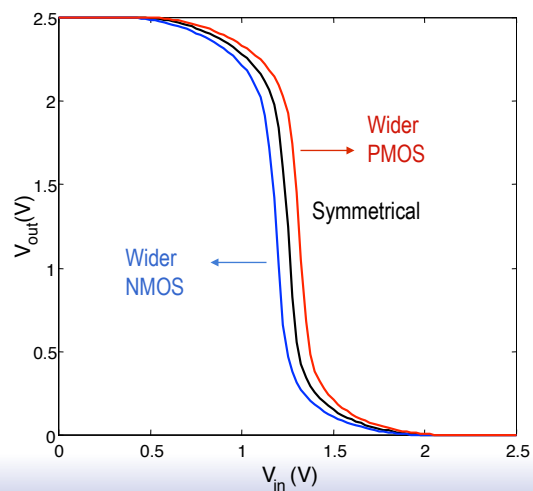


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Impact of Sizing

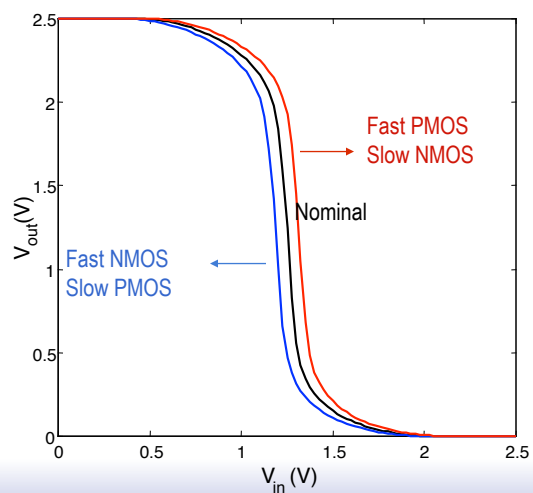


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Impact of Process Variations



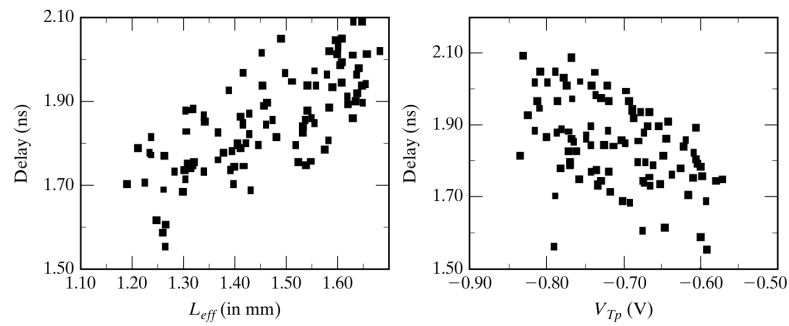
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Process Variations

Not all transistors are alike
Impacts parameters such as reliability and performance



Define process corners: SS, FF, SF, FS

CMOS Switching Delay

